

Course: Artificial Intelligence and Data Analysis

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This is a re-done project for the “Data Analysis” course.

Project overview

For this project, I worked with a sales dataset in CSV format and built a very simple analysis around it. The CSV file itself is a random example dataset file I found online.

The python script reads the CSV file, then designs a 2-layer donut pie chart that shows the distribution of profit between the different countries. The countries are arranged in the outer layer of the graph and are grouped by the regions, which are shown on the inner layer of the graph.

The script is also interactive and provides a “highlighted” view of each country and region when hovering the mouse over it - showing small-written info in larger fonts and colors for ease of view.

Main work done

All the work on this project was done in Visual Studio.

I started by designing a framework in simple language showing what I want to achieve, what I need, and then slowly crafting the lines of code to meet the requirements.

There were a few attempts in different forms/approaches (I initially planned to show several graphs with a small side-menu using Streamlit, but abandoned that idea as it felt like a wrong approach for my level of learning, and instead focused on logic-building), and they are marked with #-signs and explanations in the code body itself.

Outcome(s)

I produced a simple but interactive graph based on a file.

All of the parameters are adjustable (colors, fonts, distances, etc.) and easily findable.

I learned about the different types of variables and objects that are used in python (in my project it was the difference between “wedges” and “texts”), the construction of ‘preparation variables’ and ‘preparation functions’ (blocks of code that are designed ahead of use and not in-use, for ease of use).